

1 HI-Jass User Manual

1.1 Running the application

1.1.1 Linux

`./run_hi_jass.sh`

1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

`.\run_hi_jass.bat`

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

1.2 Default NBI configuration

The default configuration contains two neutral beams:

Beam	Species	Injection energy	Injected power	Tangent R_t	Direction
NBI-1	D	120 keV	10 MW	0.5 m	co
NBI-2	T	180 keV	0.1 MW	0.6 m	counter

1.3 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam j :

$$E_{c,j} = \left(\frac{m_j}{m_e} \right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8 \frac{A_j T_e [\text{eV}]^{3/2}}{n_e [\text{cm}^{-3}] \ln \Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3} \ln \left[1 + \left(\frac{E_{b,j}}{E_{c,j}} \right)^{3/2} \right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j} \tau_{S,j}}{E_{b,j} (10^3 e) V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

1.4 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}, \quad 0 \leq \rho \leq 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point (R_t, Z_t) in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b) \int_{\text{chord}} n_e(\rho(s)) ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \quad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With $p_n = 0$, the density is uniform along the computed chord; with $p_n > 0$, the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

1.4.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere:** the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki:** the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and Z_{eff} dependence. The fit is published for $100 \leq E \leq 10^4$ keV/u, $10^{12} \leq n_e \leq 10^{15}$ cm⁻³ and $1 \leq T_e \leq 50$ keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual:** the entered manual shine-through fraction is used directly; captured power is $(1 - f_{\text{shine}})P_{NB}$.

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

1.5 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Losses section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Losses section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect** ($q_* \rho_{Li}$) the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary A)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.

1.5.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed $R_t = R_0$, length $2a$) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0} \sigma_{\text{stop}}(E_{b,j}/A_j) e^{-n_{e0} \sigma_{\text{stop}} x}, \quad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so $\rho = 0$ at the chord mid-point and $\rho = 1$ at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

Off-axis / vertically-shifted injection. When a beam's tangency radius R_t differs from R_0 , or its aim height $Z_t \neq 0$, the birth profile is built instead from the *real* chord: a horizontal line tangent to the cylinder $R = R_t$ at height Z_t , sampled through elliptical flux surfaces

$$\rho(s)^2 = \left(\frac{R(s) - R_0}{a} \right)^2 + \left(\frac{Z_t}{\kappa a} \right)^2, \quad R(s) = \sqrt{R_t^2 + (s - s_0)^2},$$

with the chord length running edge-to-edge and ρ no longer reaching 0 (the beam no longer passes through the magnetic axis). This is the same geometry the shine-through / capture calculation already uses. For the standard on-axis tangential case ($Z_t = 0$, $R_t = R_0$ or unset) the crude $\rho(x) = |x - a|/a$ mapping above is kept, so existing results are unchanged.

Central column. The beam enters from the outboard side. If its tangency radius is inside the central column ($R_t < R_{\text{post}}$, set by **Centre-post** R; default $R_0 - a$), the straight chord would strike the column, so only the **outboard leg** from the outboard-edge entry to the point where $R(s) = R_{\text{post}}$ is kept; the (unreachable) far leg is discarded. A beam with R_t at or outboard of R_{post} threads the gap between the column and the inboard plasma edge and keeps its full chord. A beam whose aim point lies outside the plasma altogether contributes no first-orbit loss. The Deposition tab's geometry panel draws the column and marks a blocked beam.

1.5.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_* \rho_{Li,j}, \quad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}, \quad \rho_{Li,j} = \frac{m_j v_{b,j}}{e B_t}.$$

q_* is the cylindrical, elongation-corrected safety factor and $\rho_{Li,j}$ the toroidal-field gyroradius at $E_{b,j}$.

1.5.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\text{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \quad (\text{co / counter}),$$

and an ion born at x is counted as lost if $\rho_{\text{eff}}(x) > 1$. Because $\rho(x) \leq 1$, the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with $\rho(x) > 1 - \Delta r_j/a$.

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

1.5.4 Loss fraction and power

$$f_{\text{orbit},j} = \frac{\int_{\text{lost}}^x \dot{n}(x) dx}{\int_0^{2a} \dot{n}(x) dx}, \quad P_{\text{orbit},j} = f_{\text{orbit},j} P_{\text{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives T_e , T_i and $n_{b0,j}$. Injection direction (co-/counter-current) is set per NBI in its input section.

1.5.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam, $\rho_{Li,j}$ and Δr_j (in cm and as fractions of a) and $f_{\text{orbit},j}$.

1.5.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label $\rho(x) = |x - a|/a$ folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular / elliptical cross-section, large-aspect-ratio orbit widths (the birth chord itself now honours R_t and Z_t , but the orbit-width and pitch estimates are still evaluated at R_0); Δr_j and $\rho_{Li,j}$ are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density n_{e0} , not the peaked profile used elsewhere.

- When ρ_{Li}/a or $\Delta r/a \gtrsim 0.3$ orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

1.5.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an $O(a)$ fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p} \rho_{Li,j}, \quad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a \sqrt{(1 + \kappa^2)/2}},$$

(not $q_* \rho_{Li}$, whose q_* runs away as $\varepsilon \rightarrow 1$), together with the circulating and banana widths and the arbitrary- A trapped fraction

$$w_{\text{pass},j} = \varepsilon \rho_{\theta,j}, \quad w_{\text{ban},j} = 2\sqrt{\varepsilon} \rho_{\theta,j}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}},$$

with $\varepsilon = a/R_0$ (clamped to $[10^{-3}, 0.95]$) and $f_c = 1 - f_t$ (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary- A edge safety factor q_a (Uckan / ITER Physics Basis form, with the $(1.17 - 0.65\varepsilon)/(1 - \varepsilon^2)^2$ shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel uses the same Beer-Lambert deposition density as the large-aspect model and the Deposition tab's chord plot, $n_e \sigma e^{-n_e \sigma x}$ via the selected stopping model so it responds to n_{e0} the same way everywhere in the code (an earlier version used a fixed "penetration rule of thumb" mean free path that saturated at its clamp for realistic densities, decoupling the ST models' loss fraction from n_{e0} ; this has been fixed).

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ($v_{\parallel}$ along I_p), outward for counter with a weaker shift $c_t w_{\text{pass}}$ ($c_t = 0.25$) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ($\rho > 1 - \rho_{Li}/a$, born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

Mean-shift variant. Deposition-weighted loss integrals over $\rho(x)$, blended by f_c, f_t (sign convention: $-$ co, $+$ counter):

$$f_{\text{orbit},j} = f_c L_{\text{pass}} + f_t L_{\text{trap}},$$

$$L_{\text{pass}} : \rho \mp \frac{w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\text{trap}} : \rho + \frac{w_{\text{ban},j}}{2a} \mp \frac{0.25 w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

Pitch-resolved variant. A 2-D integral over birth radius x and the pitch **magnitude** $|\lambda| = |v_{\parallel}/v|$. The birth value $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x - a)^2}$ is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of $v_{\parallel}$ relative to I_p flips (co

$\leftrightarrow$ counter), carried by $s_{\text{dir}} = \mp 1$. $|\lambda_0|$ is smeared by a $\sigma = 0.10$ Gaussian. An ion is trapped if $|\lambda| < \sqrt{2\varepsilon_{\text{loc}}/(1 + \varepsilon_{\text{loc}})}$ ($\varepsilon_{\text{loc}} = \rho a/R_0$); its outboard radial excursion is

$$\text{passing : } s_{\text{dir}} \frac{w_{\text{pass},j}}{a} |\lambda|, \quad \text{trapped : } \frac{w_{\text{ban},j}}{2a} \frac{|\lambda|}{\lambda_{\text{tr}}} + s_{\text{dir}} \frac{0.25 w_{\text{pass},j}}{a},$$

(the banana half-width $\rightarrow 0$ for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if $\rho + (\text{excursion}) > 1$ or $\rho > 1 - \rho_{Li,j}/a$.

Both variants collapse toward the large-aspect co-current zero as $\varepsilon \rightarrow 0$. They remain 0-D order-of-magnitude estimates absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model's (typically a factor ~ 1.5 , not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI, $A \sim 1.4$); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

1.6 Charge-exchange loss

A captured fast ion can charge-exchange with a background cold neutral during slowing-down, becoming a fast **neutral** that escapes the plasma before fully thermalizing a second loss channel, applied to the post-orbit-loss captured power (Step 0, same ordering as first-orbit loss above). Design note: docs/CX model.pdf. A **CX-loss model** selector (rail, "Losses" section) picks between three modes; only the active mode's inputs matter.

1.6.1 Manual fraction (default)

The original flat efficiency knob `cx_loss_fraction` $\in [0, 1]$: a fixed fraction of captured power is removed, independent of E_b , n_e , or any background-neutral density. PPPL-1280 Sect. 6.2.2/6.2.3 quotes "up to 10%" of injected power for $Z_{\text{eff}} = 1$ as a rough qualitative figure, not a formula. Default 0=off; leaves every other result byte-identical.

1.6.2 Manual n_0/n_e the physics-based survival integral

The user supplies n_0/n_e directly (a spatially uniform, 0-D background neutral density $n_0 = (n_0/n_e)n_{e0}$), and the model evaluates the actual CX survival integral along the slowing-down path. For a fast ion at energy E :

$$\nu_{\text{cx}}(E) = n_0 \sigma_{\text{cx}}(E) v(E), \quad S(E) = \exp \left[- \int_E^{E_b} \frac{\nu_{\text{cx}}(E')}{|dE'/dt|} dE' \right],$$

with particle- and power-loss fractions

$$f_{\text{cx},N} = 1 - S(0^+), \quad f_{\text{cx},P} = \int_0^{E_b} \frac{E}{E_b} \frac{\nu_{\text{cx}}(E)}{|dE/dt|} S(E) dE.$$

$f_{\text{cx},P}$ (the quantity actually applied to P_{useful}) is smaller than $f_{\text{cx},N}$ because σ_{cx} falls steeply above ~ 30 to 50 keV/amu, so most CX events happen late in the cascade, when little energy is left to remove. $1/|dE/dt| = (\tau_{se}/2)\sqrt{E}/(E^{3/2} + E_c^{3/2})$ is exactly the same slowing-down kernel already used by `thermalization_time/slowing_down_distribution` (differentiating $\tau_s(E) = (\tau_{se}/3) \ln[1 + (E/E_c)^{3/2}]$ reproduces it), so no new slowing-down physics is introduced only the extra $\nu_{\text{cx}}(E)$ factor under the same integral. $\sigma_{\text{cx}}(E)$ is the Janev & Smith (1993, Sect. 2.3.1)

analytic form, refit by Swaczyna, Bzowski & Kubiak (arXiv:2411.13174, 2024, their Eq. A1) against Schultz *et al.* (2023) theory and the Barnett (1990) tabulation quoted accuracy 5% from 1100 keV/amu, the range this project’s beams live in.

Also reported, per beam, on the Dashboard: the dimensionless figure of merit $\gamma_{\text{cx}} \equiv \nu_{\text{cx}}(E_b) \tau_s \approx n_0 \sigma_{\text{cx}}(E_b) v_b \tau_s$ negligible for $\gamma_{\text{cx}} \ll 1$, worth taking seriously for $\gamma_{\text{cx}} \sim 0.1$ to 1, and a sign the single-pass loss treatment is breaking down for $\gamma_{\text{cx}} \gtrsim 1$ (flagged with a warning, as is $f_{\text{cx},P} > 10\%$, PPPL-1280’s rough $Z_{\text{eff}} = 1$ ceiling). Because CX loss depends on T_e (via $E_c(T_e)$) while T_e itself depends on the CX-reduced P_{useful} , this is a genuine fixed point; an outer under-relaxed Picard loop (in `hotjass_core`, mirroring the existing alpha-heating fixed point) closes it, keeping the core solver a single non-iterative solve.

1.6.3 Penetration (n0_LCFS/ne) edge-penetration estimate

Instead of a single uniform n_0 , the boundary value $n_{0,\text{LCFS}} = (n_{0,\text{LCFS}}/n_e) n_{e,\text{LCFS}}$ is set from the user ratio, and the core profile follows the standard penetration/recycling picture. $n_{e,\text{LCFS}}$ is evaluated at $\rho = 0.95$, not literally at $\rho = 1$: the core density shape $n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}$ is an exact zero at $\rho = 1$ whenever $p_n > 0$, which would make $n_{0,\text{LCFS}} = 0$ regardless of the input ratio – a knob that does nothing. $\rho = 0.95$ is a compromise: with a strongly peaked profile ($p_n \gtrsim 1$ to 2) it can still sit orders of magnitude below n_{e0} , so treat this mode as unreliable for very peaked density profiles.

$$\lambda_0(\rho) = \frac{v_0}{n_e(\rho) [\langle \sigma v \rangle_{\text{ion}}(T_e) + \langle \sigma v \rangle_{\text{cx}}(T_i)]}, \quad n_0(\rho) \approx n_{0,\text{LCFS}} \exp \left[- \int_{\rho}^1 \frac{a}{\lambda_0(\rho')} d\rho' \right],$$

volume-averaged to a single $\langle n_0 \rangle$ and fed through the same survival integral as ”Manual n0/ne” above. v_0 is a round ~ 3 eV Franck-Condon-like dissociation speed. **Caveat:** the ionization ($\langle \sigma v \rangle_{\text{ion}}(T_e)$) and thermal-CX ($\langle \sigma v \rangle_{\text{cx}}(T_i)$) rate coefficients used for λ_0 are an interim hand-built approximation (a threshold-shaped form for ionization, a threshold-free $\sqrt{T}$ form for CX, each capped to saturate at a few $\times 10^{-14}$ m³/s) a calibrated Voronov (1997) / ADAS `adf11` fit was not sourced/verified in this session, so treat $n_0(\rho)$ ’s absolute scale as order-of-magnitude only; the survival-integral physics downstream of $\langle n_0 \rangle$ (same $\sigma_{\text{cx}}(E)$ fit as ”Manual n0/ne”) is not affected by this caveat. $n_0(\rho)$ itself (per the active mode – flat for ”manual n0/ne”, the profile above for ”penetration”, nothing for ”manual fraction”) is plotted on the Profiles tab; because λ_0 is typically much smaller than a , the profile can span many orders of magnitude edge-to-core (shown on a log axis) – a genuine, if extreme, feature of this reduced model, not a numerical artifact.

1.6.4 Escape probability (informational only)

A fast neutral born from a CX event can re-ionize before reaching the wall, so treating every CX event as a full loss is a conservative upper bound, not exact. An escape-probability factor $\exp(-x_{\text{wall}}/\lambda_{\text{reion}})$ is shown per beam (Dashboard) $x_{\text{wall}} \approx a$, λ_{reion} from the same combined ionization+CX stopping cross section a beam neutral itself uses but is **not** applied to $f_{\text{cx},P}$; it is informational, per the design note’s explicit ”acceptable as the conservative default, but shown as such.”

1.7 Bulk toroidal rotation

A **Rotation model** selector (Losses section, below the CX-loss fields) picks between three modes, off by default – like every other loss/model knob, this leaves every existing result byte-identical

unless explicitly enabled.

Where rotation matters and where it doesn't. v_ϕ (typically $\sim 10^2$ to 10^3 km/s) is small next to the beam's own velocity (~ 3 to 5×10^3 km/s for a 100180 keV D/T beam), so shine-through, first-orbit loss and CX loss – all governed by the beam's own velocity – are essentially insensitive to it and are **not** touched. The one channel where it matters cleanly is **beam-target fusion**: that reaction's cross-section and flux factor depend on the beam-target *relative* velocity, which the un-rotated model implicitly assumes equals the beam's lab velocity (a stationary target). With rotation:

$$v_{\text{rel}}(E) = |v_{\text{beam}}(E) - s v_\phi|, \quad s = +1 \text{ (co-current beam), } -1 \text{ (counter),}$$

used for *both* the flux factor and the cross-section's energy argument (re-expressed as the equivalent deuteron energy $E_D = \frac{1}{2} M_D v_{\text{rel}}^2$, the same conversion `beam_target_reactivity_spectrum()` already used for a T beam). $v_\phi = 0$ reproduces the original stationary-target result exactly.

Non-monotonic, not just a small correction in one direction. The D-T cross-section peaks around ~ 64 keV (deuteron-equivalent energy) and falls off on both sides. For a beam energy *above* that peak (this project's 120180 keV presets all are), pulling v_{rel} down (co-current beam, co-rotating plasma) moves the *high-energy tail* of the slowing-down spectrum back toward the peak and can *increase* beam-target yield, while a counter-current beam sees the opposite – the sign is not "co always suppresses fusion", it depends on where E_b sits relative to the cross-section peak, and the net effect is an integral over the whole slowing-down spectrum (not just E_b itself), evaluated numerically.

1.7.1 Manual v_ϕ

Direct input of a single, spatially-uniform v_ϕ (m/s, positive = co-current) – the honest option when the collisionality of the source is unknown but a plausible rotation speed is.

1.7.2 Momentum balance

v_ϕ instead comes from a 0-D toroidal angular-momentum balance, the rotation analogue of this project's energy balance:

$$T_{\text{NBI}} = \sum_j s_j \frac{2P_{\text{useful},j} R_{t,j}}{v_{b,j}}, \quad v_\phi = \frac{T_{\text{NBI}} \tau_\phi}{\rho_i R_0 V},$$

summing each beam's injected canonical angular momentum ($s_j = +1$ co, -1 counter; $R_{t,j}$ its own tangency radius), balanced against a momentum-confinement loss L/τ_ϕ with $L \sim \rho_i v_\phi R_0 V$ (ρ_i = thermal-ion mass density, $n_{D0} M_D + n_{T0} M_T$) – the bulk plasma treated as a single representative rotating mass at the magnetic axis, the same level of 0-D reduction as everywhere else in this model. **Caveat:** unlike τ_E , there is no widely validated scaling for the momentum confinement time τ_ϕ ; the user-set ratio `tau_phi / tauE,i` (default 1.0, the common crude $\tau_\phi \approx \tau_{E,i}$ assumption) sets $\tau_\phi = (\text{ratio}) \times \tau_{E,i}$, so treat v_ϕ 's absolute scale here as order-of-magnitude, more so than the manual-input mode.

The Dashboard reports v_ϕ (km/s) and a Mach number $M = v_\phi/c_s$ ($c_s = \sqrt{(T_{e0} + T_{i0})/m_i}$, a simple ion-acoustic estimate, not a rigorous sound speed), plus the net torque T_{NBI} for the momentum-balance mode.

1.8 Beam-beam fusion

Off by default (checkbox, Losses section, below the rotation fields). Fast ions from **two separate NBI sources** can fuse with each other directly – a third fusion channel alongside thermal-thermal and beam-target. This is a **reduced** model, not a full treatment: instead of integrating over both beams’ slowing-down spectra (a genuine 2-D relative-velocity integral), each population is approximated by a single characteristic velocity at its own average fast-ion energy (`average_fast_energy_keV`, the same representative energy this project already uses for pressure/energy-density diagnostics), signed by its own co/counter direction. Only the **first two** useful beams are paired – this project’s tested scope is exactly two NBI sources.

The relevant physical quantity is the beam-beam **relative** velocity:

$$v_{\text{rel}} = |s_1 v_1 - s_2 v_2|, \quad s = +1 \text{ (co)}, -1 \text{ (counter)},$$

fed into the same deuteron-equivalent-energy cross-section evaluation as the rotation correction to beam-target fusion (Sec. “Bulk toroidal rotation” above) – structurally the same construction, with the *other beam’s own velocity* playing the role v_ϕ played there. **This is why counter-injecting the second beam matters physically:** two co-current beams at similar per-amu energy have a small v_{rel} (their velocities nearly cancel) and thus a negligible beam-beam yield, while opposite directions maximize v_{rel} – exactly DANTE’s default configuration (D co-current, T counter-current). D-T and D-D pairs are handled (the only cross-section fits this project carries); T-T pairs, or a beam-beam calculation with fewer than two useful beams, contribute nothing.

The result is folded into the existing D-T/D-D fusion power and neutron- rate totals (`pf_dt_w`, `pf_dd_w`, `neutron_rate_s`) exactly like the thermal and beam-target contributions, so it is automatically included in the alpha-heating closure (Sec. “Alpha self-heating” below) with no separate wiring needed. The Dashboard’s D-T/D-D fusion-power rows show all three contributions (thermal / beam-target / beam-beam); the Scan tab’s “Neutrons” group breaks the neutron rate out the same way, with the beam-beam panel showing a genuine (if usually small, for DANTE’s default 0.1 MW counter-beam) nonzero curve once enabled, rather than the flat zero it shows while off.

1.9 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for T_e and T_i . The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

1.9.1 Beam heating split

Each useful beam j delivers a fraction $L_e(x_j)$ of its power to electrons and $L_i(x_j) = 1 - L_e(x_j)$ to ions, with $x_j = E_{b,j}/E_{c,j}(T_e)$ and

$$L_i(x) = \frac{1}{x} \int_0^x \frac{du}{1 + u^{3/2}}, \quad L_e(x) = 1 - L_i(x).$$

$L_i \rightarrow 1$ as $x \rightarrow 0$ (cold, ion-dominated) and $L_i \rightarrow 0$ as $x \rightarrow \infty$ (fast beam, electron-dominated). $P_{\text{useful},j}$ is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

1.9.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \quad P_{\text{aux},i} = (1 - f_e^{\text{EC}}) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant, f_e^{EC} close to 1. For ICRH the electron and ion fractions are set independently; if $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$ the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

1.9.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e \left(\frac{E_\alpha}{E_c^\alpha(T_e)} \right), \quad E_c^\alpha = \left(\frac{m_\alpha}{m_e} \right)^{1/3} T_e,$$

so $P_{\alpha,e} = L_e^\alpha P_\alpha$ and $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$. Because P_α depends on $P_{f,\text{tot}}$, which depends on T_i , which depends on $P_{\alpha,i}$, the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

1.9.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for T_e :

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$ is the thermal-ion density left after the fast ions are subtracted. A non-positive n_{thermal} marks the operating point infeasible.

1.9.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power $P_{ei}(T_e, T_i)$ (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\frac{3}{2}n_{e0}T_e(10^3e)\frac{V}{\tau_{E,e}} = \sum_j L_e(x_j)P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i),$$

$$\frac{3}{2}n_{\text{thermal}}T_i(10^3e)\frac{V}{\tau_{E,i}} = \sum_j L_i(x_j)P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i).$$

The reported P_e and P_i remain the pure beam split; P_{ei} , P_{α} and P_{aux} are reported separately.

1.9.6 Profile-corrected 0-D

The balance above is genuinely 0-D: one density, one T_e , one T_i , treated as uniform over the volume. With the **profile-corrected 0-D** checkbox (Plasma section) *off*, the entered central density is used directly as that uniform value and the solved T is a flat-plasma effective temperature; comparing it to a measured on-axis T_0 then needs a peaking factor supplied by the reader (validation runs use $\simeq 2$). In this mode every radial integral (shine-through column density, thermal and beam-target fusion, pressure, thermal energy) also uses **flat** profiles, consistent with the balance: the peaking parameters are ignored. (Before October 2026 these integrals still applied the $(1 - \rho^2)^{2p}$ shapes and treated the flat values as on-axis peaks, which with $p_{Te} = 1$ cut the profile-average T to a third and under-reported fusion power by up to $\sim 3\times$.)

With the checkbox *on*, the entered **central n_e** is taken to be the **on-axis** n_{e0} , and for a $(1 - \rho^2)^{2p}$ profile the volume average is $\langle X \rangle = X_0/(1 + 2p)$. The energy balance and the τ_E scalings then run on

$$\langle n_e \rangle = \frac{n_{e0}}{1 + 2p_n},$$

so the solved T_e , T_i are **volume-averaged**. The electron and ion temperature profiles are shaped by **independent** exponents p_{Te} (**Temp peaking (electron)**) and p_{Ti} (**Temp peaking (ion)**); a value < 0 means "use p_{Te} "), so the on-axis values are

$$T_{e0} = \langle T_e \rangle (1 + 2p_{Te}), \quad T_{i0} = \langle T_i \rangle (1 + 2p_{Ti}).$$

They are reported (Dashboard row "on-axis T_{e0}/T_{i0} ") and are what the fusion, pressure and β_t integrals use as their central values the reactivity integrals with the $(1 - \rho^2)^{2p_{Ti}}$ ion shape, the beam slowing-down kernel with the $(1 - \rho^2)^{2p_{Te}}$ electron shape. Set p_n (**Density peaking**), p_{Te} and p_{Ti} to match the real profiles; $p \simeq 0.5$ to 0.75 gives $X_0/\langle X \rangle \simeq 2$ to 2.5 (L-/H-mode), while a **hot-ion mode** on a compact ST needs $p_{Ti} \gg p_{Te}$ ($T_{i0}/\langle T_i \rangle \simeq 3$ to 4 with T_e nearly flat). With all peaking parameters at 0 the correction is a no-op.

1.9.7 Deposition-weighted fast-ion profile

With profile-corrected 0-D on, the fast-ion density is built **locally** from the beam deposition (October 2026) instead of a single 0-D value spread like the slowing-down time:

$$n_b(\rho) = \sum_j S_j h_j(\rho) \tau_{s,j}(n_e(\rho), T_e(\rho)), \quad S_j = \frac{P_{\text{useful},j}}{E_{b,j} e},$$

with $\tau_{s,j}$ the local thermalisation time and $h_j(\rho)$ the confined birth rate per unit volume of beam j , normalised so that $V\langle h_j \rangle = 1$ (**physics.beam.birth_density_profile**):

- **Attenuation** along tangential chords (Beer-Lambert) on the $n_{e0}(1 - \rho^2)^{2p_n}$ profile, with the same stopping cross-section as the shine-through model.
- **Finite beam width.** A pencil chord tangent at R_t deposits a finite number of ions into the vanishing volume of the flux surface $\rho(R_t)$, so its birth density per volume diverges there. The beam is therefore sampled as 5×5 parallel sub-chords over a Gaussian cross-section $\exp(-r^2/w^2)$ with $w = 0.10$ m, shifted in tangency radius and height.
- **Prompt loss** (already counted in the orbit-loss power), two options (`orbit_loss_deposition`; HOT-Jass: MODELS, "Orbit loss consistent in deposition"):
- **"cutoff"** (default, historical): every birth beyond `orbit_cutoff_rho` is removed. That radius is set by the worst-case (trapped) orbit, so at low current, where the banana width approaches the minor radius, it removes far more births than the loss formula removes power (TCV at 150 kA: 72 % of births against 13 % of power), and the confined beam is renormalised onto the remaining core births.
- **"consistent"** (ST orbit models): each birth is weighted by $1 - P_{lost}(\rho, |\lambda_0|)$, the per-birth rule of the pitch-resolved loss formula (`physics.st_orbit_loss_probability`, also used for the lost power), with $|\lambda_0| = R_t/R$ along each chord. It changes NBCD by about 30 % on TCV, about 6 % on MAST and 6-19 % on NSTX-U, and is therefore an option, off by default.
- **Orbit-width smoothing.** Births are spread over half the passing orbit width (at least 0.03 in ρ), as a *volume-weighted* average of the birth density, so that births do not pile up on the axis.

The volume average of $n_b(\rho)$ replaces the 0-D n_{b0} inside the balance, so dilution, the thermal density $n_{th}(\rho) = n_\Sigma(\rho) - n_b(\rho)$ (locally depleted where the fast ions sit) and the thermal and beam-target fusion integrals all use the same fast ions. The solver stores the radial profiles on `OperatingPoint` (`rho_profile`, `nb_profile_m3`, `n_fuel_profile_m3`, `ne_profile_m3`, `Te_profile_keV`, `Ti_profile_keV`, the fusion power densities, and per beam the birth rates $S_j h_j(\rho)$ and birth pitch $\xi_{0,j}(\rho)$). Their volume integrals are exactly the reported totals. With profile-corrected 0-D off the flat model is kept unchanged.

On the ASCOT5/BBNBI DANTE reference case (4 MW D at 80 keV) the deposition model gives 1.24 MW of fusion power against ASCOT's 1.4 MW; the earlier single- n_{b0} model gave 0.87 MW.

1.10 Particle balance and fuelling

The power balance fixes T_e and T_i at a given density, but says nothing about how that density is sustained. The HOT-Jass web app (**Balance** tab) adds a steady-state **particle balance** for the thermal fuel ions as a post-processing step: it uses the solved operating point and does not feed back into the solve.

1.10.1 Fast-ion dilution

At fixed n_e , quasi-neutrality makes every fast beam ion displace a thermal ion ($n_{\text{thermal}} = n_{\text{sum}} - n_{b0}$, see *Power balance*). The share of fast ions,

$$f_{\text{dil}} = \frac{n_{b0}}{n_{\text{thermal}} + n_{b0}},$$

reduces both thermal and beam-target fusion, which scale with the thermal fuel densities. It is largest for low-energy beams into a moderate-density, hot plasma (long slowing-down time), e.g. $\sim 14\%$ for the DANTE design point.

1.10.2 Steady-state balance per species

For each thermal fuel species $s = \text{D, T}$, holding the input D:T mix at the solved density,

$$S_{\text{ext},s} + S_{\text{beam} \rightarrow s} - R_{\text{burn},s} = \frac{N_s}{\tau_p^*}, \quad N_s = \langle n_s \rangle V, \quad \tau_p^* = \left(\frac{\tau_p^*}{\tau_E} \right) \tau_{E,e},$$

where

- N_s is the thermal inventory of species s (volume-average density from the balance times the plasma volume), and N_s/τ_p^* its particle loss rate;
- τ_p^* is the **effective** particle confinement time including recycling, entered as the ratio τ_p^*/τ_E (Plasma input, default 2). In a low-recycling regime $\tau_p^* \approx \tau_p$; with strong recycling $\tau_p^* \gg \tau_p$;
- $S_{\text{beam} \rightarrow s}$ is the rate at which beam ions thermalise into their own species (a D beam feeds D): $S_{\text{beam}} = \sum_j P_{\text{useful},j}/(E_{b,j} e)$, minus the fast ions burnt in beam-target reactions before they thermalise;
- $R_{\text{burn},s}$ is the thermal ions consumed by fusion: one D and one T per D-T reaction (thermal-thermal reactions consume one of each thermal species; beam-target reactions consume the fast ion and one thermal ion of the other species). D-D burn-up ($\sim 10^{-3}$ of D-T) is neglected;
- $S_{\text{ext},s}$ is the external fuelling (gas puff or pellets) needed to hold the mix. It is also given as a gas throughput, using 1 Pa m^3 of D_2/T_2 at $273 \text{ K} = 5.3 \times 10^{20}$ atoms.

1.10.3 Over-supplied species

$S_{\text{ext},s} < 0$ means the beam alone delivers more of species s than the plasma loses: the mix can then only be held by pumping. With no external fuelling of that species its thermal fraction settles at

$$f_s = \frac{(S_{\text{beam} \rightarrow s} - R_{\text{burn},s}) \tau_p^*}{N_D + N_T}.$$

This is the typical situation for D beams into a tritium-rich target (DANTE: $S_{\text{beam}}(\text{D}) = 4.85 \times 10^{20} \text{ s}^{-1}$ against a D loss of $2.85 \times 10^{20} \text{ s}^{-1}$ at D:T = 10:90, so the D fraction would rise to $\simeq 0.17$ with tritium-only fuelling). Hydrogen beams are reported separately: H is not part of the D:T mix and has to be pumped.

1.10.4 Power balance view

The same tab shows the power balance of *Power balance (electron and ion)* as two pies: heating sources ($P_{\text{NB},e}$, $P_{\text{NB},i}$, $P_{\text{aux},e}$, $P_{\text{aux},i}$, P_α) and transport losses $W_e/\tau_{E,e}$, $W_i/\tau_{E,i}$, with $W_e = \frac{3}{2} \langle n_e \rangle \langle T_e \rangle V$ and $W_i = \frac{3}{2} n_{\text{thermal}} \langle T_i \rangle V$. In steady state the two losses sum to the total heating; P_{ei} is an internal exchange, not a loss.

1.10.5 Limitations

- The result scales directly with τ_p^* , which is an input, not computed; the absolute fuelling rates are only as good as that choice. The sign of $S_{\text{ext},s}$ (whether the beam over-supplies a species) is far less sensitive.
- D and T are assumed to share one τ_p^* ; isotope effects on particle transport and wall recycling are not modelled.
- Volume-averaged (0-D) balance: no fuelling-depth or density-profile effects (pellet versus gas puff).

1.11 Non-inductive current (NBCD and bootstrap)

The HOT-Jass web app (**Current** tab) estimates the neutral-beam driven and the bootstrap current of a solved operating point (`hotjass.current`, `HotJassModel.current_drive(op)`). It is a **diagnostic only**: nothing feeds back into the solve, so an operating point whose non-inductive current exceeds I_p is flagged rather than corrected. It needs profile-corrected 0-D, because it uses the solver's radial profiles, the beam deposition and the birth pitch; with flat profiles there are no gradients and no current is computed.

1.11.1 Neutral-beam current drive

The fast-ion current of each beam is the local 1-D Cordey / Start-Cordey result for a mono-energetic source, with electron and ion drag and ion pitch-angle scattering (no energy diffusion):

$$j_f = Z_b e S h(\rho) \tau_s v_b \xi_0 J(u_c, \beta), \quad J = \int_0^1 \frac{u^3}{u^3 + u_c^3} \left[\frac{u^3(1 + u_c^3)}{u^3 + u_c^3} \right]^{\beta/3} du,$$

where:

- $S h(\rho)$ is the confined birth rate per volume (above), and τ_s the Spitzer slowing-down time ($dv/dt = -(v/\tau_s)(1 + v^3/v_c^3)$).
- $\xi_0 = R_t/R$ is the birth pitch. A horizontal chord tangent at R_t crosses radius R with toroidal direction cosine R_t/R ; it is averaged over the same sub-chords as h and weighted by births, and is negative for counter-current beams.
- $u = v/v_b$, $u_c^3 = (E_c/E_b)^{3/2}$, with the critical energy from the local composition (D, T and the carbon-like impurity), $E_c = 14.8 A_b T_e (\sum_j n_j Z_j^2 / n_e A_j)^{2/3}$.
- $\beta = Z_{\text{eff}}/\bar{Z}$, with $\bar{Z} = \sum_j n_j Z_j^2 (m_b/m_j)/n_e$, is the ratio of the ion pitch-angle scattering rate to the ion drag on the speed. (Written as $l(l+1)\beta'/3$ with $\beta' = Z_{\text{eff}}/2\bar{Z}$ elsewhere in the literature; for the $l = 1$ moment both give the exponent $Z_{\text{eff}}/3\bar{Z}$.)

The electron shielding with the trapped-electron correction (Start & Cordey 1980) gives the net driven current:

$$j_{NB} = j_f \left[1 - \frac{Z_b}{Z_{\text{eff}}} (1 - G) \right], \quad G = \left(1.55 + \frac{0.85}{Z_{\text{eff}}} \right) \sqrt{\epsilon} - \left(0.2 + \frac{1.55}{Z_{\text{eff}}} \right) \epsilon,$$

with $\epsilon = \rho a/R_0$ and G clipped to $[0, 1]$. Total currents and the current-drive efficiency:

$$I = \frac{V}{2\pi R_0} \langle j \rangle, \quad \eta_{CD} = \frac{I_{NB} R_0 \bar{n}_e}{P_{NB}} \quad [10^{20} \text{ A W}^{-1} \text{ m}^{-2}],$$

with P_{NB} the injected power and $\bar{n}_e$ the line-average density.

1.11.2 Bootstrap current

Sauter, Angioni & Lin-Liu, *Phys. Plasmas* **6** (1999) 2834 (erratum **9** (2002) 5140), valid at arbitrary aspect ratio and collisionality, in the cylindrical form

$$j_{BS} = -\frac{1}{B_p} \left[L_{31} \frac{dp}{dr} + L_{32} n_e \frac{dT_e}{dr} + L_{34} \alpha n_i \frac{dT_i}{dr} \right],$$

with p the **thermal** pressure (fast ions excluded; their current is the NBCD), $L_{31}, L_{32}, L_{34}, \alpha$ the Sauter fits in the trapped fraction f_t (Lin-Liu & Miller), ν_e^*, ν_i^* and Z_{eff} , and $dr = a d\rho$. The profile gradients are taken analytically from the $(1 - \rho^2)^{2p}$ shapes and the current is integrated on a fine grid clustered at the edge: for low peaking exponents ($2p < 1$) the gradients diverge at $\rho = 1$, and a finite-difference gradient would make the edge contribution grid-dependent.

The 0-D model has no current profile, so B_p and q come from an **assumed** one:

$$j \propto (1 - \rho^2)^\nu, \quad \nu = \frac{q_{\text{cyl}}}{q_0} - 1, \quad B_p = \frac{\mu_0 I(\rho)}{L_p}, \quad L_p = 2\pi\rho a \sqrt{\frac{1 + \kappa^2}{2}},$$

with $I(\rho) = I_p[1 - (1 - \rho^2)^{\nu+1}]$, q_{cyl} the cylindrical edge q (**safety_factor_cyl_edge**) and q_0 an input (Plasma row **q0**, default 1). The low-aspect-ratio toroidal enhancement of q (q_{95}/q_{cyl} , large on an ST) is not a current-peaking effect, so it enters $q(\rho)$ only, used in the collisionalities:

$$q(\rho) = q_{\text{cyl}}(\rho) \left[1 + \left(\frac{q_{95}}{q_{\text{cyl}}} - 1 \right) \rho^2 \right].$$

(Taking ν from q_{95} instead gave $\nu \approx 16$ on MAST-U and NSTX-U, an unrealistically peaked current.)

The inductive remainder is $I_{\text{ind}} = I_p - I_{\text{NB}} - I_{\text{BS}}$ (negative: over-drive), and $f_{\text{NI}} = (I_{\text{NB}} + I_{\text{BS}})/I_p$.

1.11.3 Validation

TRANSP, ASTRA and CRONOS references (details, figures and scripts in **docs/Validation/HI-Jass_validation.md**, sec. F-H). NSTX-U, MAST and TCV use the centre-of-mass cross-sections and are given for both deposition options (cutoff / consistent), with τ_E tuned to the reference stored energy or temperatures; ITER and JET at matched β_N (cutoff, earlier runs):

Case	I_{NB} model / ref.	I_{BS} model / ref.
NSTX-U 0.87 MA, 10.2 MW 80 kV, H98=1	353 / 285 / 287 kA	594 / 621 / 583 kA
NSTX-U 0.975 MA, 12.6 MW 90 kV, H98=1	404 / 327 / 370 kA	653 / 682 / 604 kA
NSTX-U 1.1 MA, 15.6 MW 100 kV, H98=1	466 / 381 / 396 kA	762 / 793 / 704 kA
NSTX-U 0.635 MA, 0.75 T, 6.8 MW 80 kV	199 / 211 / 184 kA	441 / 444 / 451 kA
MAST #18808 0.62 MA, 3.5 MW off-axis	287 / 268 / 253 kA	147-161 / ≈ 45 kA
TCV #59429 0.15 MA, 1.15 MW (ASTRA)	91-124 / 67-89 / ≈ 49 kA	74-80 / ≈ 81 kA
ITER hybrid 12 MA, β_N 2.65	3.33 / 3.5 MA	5.82 / 4.7 MA
ITER steady state (no ITB) 10 MA, β_N 2.70	4.31 / 3.6 MA	5.45 / 5.0 MA
JET #58323 1.4 MA, β_N 2.8	0.55 / 0.33 MA	0.51 / 0.32 MA

References: NSTX-U, Gerhardt et al., *Nucl. Fusion* **52** (2012) 083020 (fully non-inductive TRANSP scenarios, Table 2, broad profiles; W_{tot} matched, $f_{\text{NI}} = 1.03$ -1.07 with the consistent deposition). MAST, Turnyanskiy et al., *Nucl. Fusion* **49** (2009) 065002 (TRANSP classical; the bootstrap value is read off a stacked plot). TCV, Piron et al., *Nucl. Fusion* **59** (2019) 096012 (ASTRA profiles, digitised and integrated, about 20 % uncertainty; ECCD not modelled). ITER, Parail et al., *Nucl. Fusion* **53** (2013) 113002 (Table 3). JET, Sips, EPS 2003 O-1.3A (TRANSP; density and beam set-up only approximately known, $a = 0.95$ m used for the diverted plasma).

With the consistent deposition NBCD agrees within about 15 % on NSTX-U and 6 % on MAST; TCV stays 1.4-1.8x high (charge-exchange loss the main uncertainty). The cutoff deposition over-predicts NBCD wherever the orbits are wide. The bootstrap current agrees within about 15 % on NSTX-U and TCV and 25 % at ITER, but is about 3x the MAST reference and high on JET.

Matching β_N instead of the stored energy raises both NSTX-U currents by 20-30 % (the codes normalise β_N differently), so cross-code comparisons should match W . Remaining systematic effects: the **full-energy-only beam** ($E/2$ and $E/3$ not modelled; on NSTX-U they lowered I_{NB} by about 15 % in an earlier test) and a first-orbit loss of 25-40 % on spherical tokamaks, well above TRANSP's few per cent.

1.11.4 Limitations

- Mono-energetic beam (no $E/2$, $E/3$), no anomalous fast-ion diffusion, no current diffusion (steady state assumed).
- Assumed current profile and $q(\rho)$; no equilibrium.
- Where fast ions make the thermal-ion density hollow (fixed n_e , large n_b in the core), the thermal bootstrap reverses near the axis (seen on DANTE).
- Low-peaking presets ($p \simeq 0.1$ to 0.25) put roughly 28 % of I_{BS} at $\rho > 0.95$: the power-law shapes act like a pedestal.

1.12 Confinement time

$\tau_{E,e}$ and $\tau_{E,i}$ in the balances above are set per channel by the *Confinement* selector in the Losses section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual **tauE,e** / **tauE,i** inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in P_{loss} (it is closed by an outer fixed-point and is not known when τ_E is evaluated). None of these depend on T_e or T_i , so each is evaluated once.

1.12.1 Confinement enhancement

Every scaling below can be multiplied by a confinement enhancement h (default 0; the MODELS slider under Confinement in the HOT-Jass web app, and the config field **tauE_enhancement** in HI-Jass):

$$\tau_E = (1 + h) \tau_{\text{scaling}}, \quad H = 1 + h.$$

$h = 0$ is the plain scaling. Fixed τ_E input and the neoclassical ion channel are not affected.

1.12.2 Fixed (input)

$\tau_{E,e}$ and $\tau_{E,i}$ are the values entered in the Losses section (under the Confinement selector), used unchanged.

1.12.3 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562 I_p^{0.93} B_t^{0.15} n_{19}^{0.41} P_{\text{loss}}^{-0.69} R_0^{1.97} \kappa_a^{0.78} \varepsilon^{0.58} M_{\text{eff}}^{0.19}$$

with I_p in MA, B_t in T, P_{loss} in MW, R_0 and a in m, $n_{19} = n_{e0}/10^{19} \text{ m}^{-3}$, $\varepsilon = a/R_0$, $\kappa_a = \kappa$, and $M_{\text{eff}} = 2x_D + 3x_T$ the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$) – like every scaling offered here, it does not split electron/ion channels.

The fit was obtained from conventional-aspect-ratio devices ($A \sim 2.5$ -4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When $A = R_0/a < 2$ the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

1.12.4 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4} I_p^{1.01} B_t^{0.70} n_e^{-0.07} P_{\text{loss}}^{-0.37}$$

with I_p in A, B_t in T, n_e in m^{-3} , P_{loss} in W and τ_E in s. Selecting *Kaye NSTX L-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

1.12.5 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9} I_p^{0.57} B_t^{1.08} n_e^{0.44} P_{\text{loss}}^{-0.73}$$

with I_p in A, B_t in T, n_e in m^{-3} , P_{loss} in W and τ_E in s. Relative to IPB98(y,2) the current dependence is much weaker ($I_p^{0.57}$ vs $I_p^{0.93}$) and the toroidal-field dependence much stronger ($B_t^{1.08}$ vs $B_t^{0.15}$), the qualitative spherical-tokamak trend; the density and heating-power exponents ($n_e^{0.44}$, $P_{\text{loss}}^{-0.73}$) are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ($A \sim 1.3$ to 1.5 dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

1.12.6 Neoclassical ion

Not currently selectable from the Confinement dropdown – v1 offers only the four modes above (Fixed, Kaye L, Kaye H, IPB98(y,2)), each applied to both channels. The neoclassical estimate below is implemented in `hotjass/solve.py` and was previously reachable via an "X e / neoclassical i" hybrid selector entry; that combination is out of scope for v1 and may return in a later version. Documented here for completeness:

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\text{neo}} \simeq \frac{q^2 \rho_i^2 \nu_{ii}}{\varepsilon^{3/2}}, \quad \tau_{E,i} = \frac{a^2}{\chi_{i,\text{neo}}} = \frac{a^2 \varepsilon^{3/2}}{q^2 \rho_i^2 \nu_{ii}},$$

with $\varepsilon = a/R_0$, the elongation-corrected cylindrical safety factor

$$q = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\text{eff}} v_{th,i}}{e B_t}, \quad v_{th,i} = \sqrt{\frac{2 T_i (10^3 e)}{m_{\text{eff}}}}, \quad m_{\text{eff}} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s \in \{D, T\}} 4.80 \times 10^{-8} \frac{n_s [\text{cm}^{-3}] \ln \Lambda}{\sqrt{\mu_s} T_i [\text{eV}]^{3/2}}, \quad \mu_s = m_s / m_p.$$

Since $\tau_{E,i} \propto T_i^{1/2}$, the ion balance becomes a 1-D fixed-point root-find for T_i rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible T_i ; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

1.12.7 Neoclassical ion (arbitrary aspect ratio)

Also not currently selectable – same v1 scope note as above.

The bare $\varepsilon^{-3/2}$ geometric factor above both diverges as $\varepsilon \rightarrow 0$ and stays bounded as $\varepsilon \rightarrow 1$, neither of which is right on a spherical tokamak. The *arbitrary A* variant (previously, selector entries ending "neoclassical *i* (*arbitrary A*)") replaces the trapped-fraction part with the Lin-Liu & Miller f_t/f_c the banana-regime structure of Helander & Sigmar's *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary- A edge safety factor q_a :

$$\chi_{i,\text{neo}} \simeq q_a^2 \rho_i^2 \nu_{ii} \frac{f_t}{f_c}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}}, \quad f_c = 1 - f_t,$$

$$q_a = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2 (1 + 2\delta^2 - 1.2\delta^3)}{2} \frac{1.17 - 0.65 \varepsilon}{(1 - \varepsilon^2)^2}.$$

f_t/f_c stays finite for every $\varepsilon < 1$ and rises steeply only as $\varepsilon \rightarrow 1$ (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the $\varepsilon^{-3/2}$ form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the $\tau_{E,i} \propto T_i^{1/2}$ fixed point amplifies it), which is exactly why both are offered run them side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

1.12.8 Known limitation: the P fed into IPB98(y,2) / Kaye NSTX

These three scalings use $P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i}$ (sec. above) NBI power *after* shine-through, first-orbit loss **and** charge-exchange loss are all subtracted. The experimental

databases these scalings were fit to use a coarser convention instead (absorbed power, i.e. shine-through-corrected only they do not subtract orbit-loss/CX-loss of the already-absorbed fast-ion population), so HI-Jass currently under-feeds P into these negative-exponent scalings and is biased toward *overestimating* τ_E whenever orbit-loss/CX-loss are significant (STs especially). Not yet fixed. See **Appendix A** for a quantitative comparison, including how this compares to the (much larger) effect of the scaling choice itself.

1.13 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad 0 < E \leq E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E) dE}{\int_0^{E_{b,j}} f_j(E) dE}$$

1.14 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\text{eff}} = \frac{\sum_j P_{\text{useful},j} \tau_{S,j}}{\sum_j P_{\text{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{fast} \rangle = \frac{\sum_j P_{\text{useful},j} \langle E_{fast} \rangle_j}{\sum_j P_{\text{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{fast}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

1.15 Plasma beta: isotropic and anisotropic fast ions

Toroidal and normalised beta use the volume-averaged pressure of thermal electrons, thermal ions and fast beam ions:

$$\beta_t = \frac{\langle p \rangle}{B_0^2/2\mu_0}, \quad \beta_N = \beta_t[\%] \frac{a B_0}{I_p[\text{MA}]}, \quad \langle p \rangle = \langle n_e T_e \rangle + \langle n_i T_i \rangle + p_{fast}.$$

The fast-ion pressure depends on the pitch distribution, which the 0-D model does not resolve. With $u_{fast} = n_b \langle E_{fast} \rangle$ the fast-ion energy density and $\xi = v_{||}/v$:

$$p_{fast} = (1 - \langle \xi^2 \rangle) u_{fast}.$$

- **Isotropic** ($\langle \xi^2 \rangle = 1/3$, the default): $p_{fast} = \frac{2}{3} u_{fast}$.
- **Anisotropic** bound ($\langle \xi^2 \rangle = 1$, all fast ions parallel): $p_{fast} = 0$.

The solver reports the first as `beta_t` (pitch set by `fast_ion_mean_pitch2`) and always computes the second as `beta_t_anisotropic`.

Both are shown (the anisotropic value in brackets). Tangential beams are born nearly parallel and isotropise only by pitch-angle scattering while slowing down, so the true value lies between the two bounds: closer to the anisotropic one for tangential injection at high T_e (slowing down faster than scattering), closer to the isotropic one for perpendicular beams. The difference matters where fast ions carry a large share of the pressure (low density, high beam power). β is a diagnostic: nothing in the solve depends on it.

1.16 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, and their on-axis values when profile-corrected 0-D is on; densities, power balance terms, orbit widths, D-T / D-D fusion power, neutron rate, Greenwald ratio, edge safety factor q_* , beta). q_* is reported using whichever formula the active orbit-loss model actually uses – the bare cylindrical `safety_factor_cyl_edge` for "large-aspect", or the arbitrary- A `safety_factor_cyl_edge_arbitrary_A` (q_a) for either ST orbit variant – so it stays consistent with the orbit widths shown just above it. Per-beam CX-loss rows show $f_{cx,P}/f_{cx,N}/\gamma_{cx}$ (plus n_0 and the informational escape probability) when a physics-based CX model is selected, or just the flat fraction for "Manual fraction" – see "Charge-exchange loss" above. The rotation row shows "off" unless a Rotation model is selected, in which case it shows v_ϕ , a Mach number, and (momentum-balance mode) the net NBI torque – see "Bulk toroidal rotation" above.
- **Deposition** (1) beam targeting geometry in the torus top view (titled with the active device), tangent to each beam's R_t with its co-/counter-current sense; (2) neutral-beam survival $I(s)/I_0$ and the fast-ion birth rate $n_e \sigma I(s)/I_0$ along the beam, annotated with each beam's shine-through percentage; (3) the resulting fast-ion birth versus normalised radius ρ , per beam and total (chord-sampled, midplane circular approximation), with the prompt first-orbit-loss zone $\rho > \rho_{cut}$ shaded and a per-beam ρ_{cut} line (from the same criteria as the orbit-loss model in use); (4) the steady-state slowing-down distribution $f(E)$ with the E_b edge marked.
- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles** $n_e(\rho)$, $T_{e,i}(\rho)$, $\tau_S(\rho)$ (local thermalization time, from the same local $n_e(\rho)$, $T_e(\rho)$ used elsewhere in this panel – log scale, since it rises sharply toward the edge as $n_e \rightarrow 0$), $n_0(\rho)$ (background-neutral density behind the active CX-loss model – see "Charge-exchange loss" above), the plasma shape, and $P_{fus}(\rho)$ (local D-T + D-D fusion power density, thermal vs. beam-plasma / beam-target, built from the same on-axis densities and $(1 - \rho^2)^{2p}$ shapes as the volume-integrated totals on the Dashboard – see "Fusion power" below).
- **Fusion** a pie chart of P_{fus} split into its six channel/reaction combinations (D-T and D-D, each thermal / beam-target / beam-beam), with MW values and percentages; slices under 0.5% of the total are hidden to keep the chart readable (still fully available in the Dashboard's D-T/D-D breakdown rows). Title also shows Q and the full neutron rate Y_n .
- **Assumptions** the full validity / fit-range read-out and any warnings.

- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

1.17 Saving and loading a run

Below the results tabs, **Export** has three buttons plus a fourth for the reverse direction:

- **Summary PNG / Summary PDF** a snapshot of the Summary tab (the fixed 3x4 scan-panel grid) as an image.
- **Run record (JSON)** every plasma/beam input, every model toggle, the full solved operating point, and the Summary sheet's scan arrays, as one JSON file (needs a completed run; disabled with a status message otherwise).
- **Load run record...** the reverse of the above: pick a previously saved run-record JSON, and every input field and toggle (confinement, orbit loss, CX-loss model, rotation, beam-beam, equipartition, alpha self-heating, profile-corrected 0-D, both NBI sections) is restored from it, then the point is re-solved automatically. Tolerant of a record written by an older or newer HI-Jass version: a field this version doesn't recognise is ignored, and a field this version expects but the record lacks simply keeps its current value instead of raising an error. Only the solved *inputs* are restored – the record's own **operating_point** / **summary** output is historical reference only, not re-applied (the fresh re-solve after loading is what actually reproduces those numbers, so loading a record from a much older HI-Jass version may reproduce a slightly different result if the underlying physics changed meanwhile).

1.18 Interface font size (accessibility)

The button next to the Operating point/Scan mode toggle, above the Run button, cycles the whole interface through four zoom levels: 100% -> 115% -> 130% -> 150% -> back to 100%. Each click rescales every rail label/entry/button, the Dashboard, and all plot text (titles, axis labels, legends, tick labels) together, redrawing the currently solved point immediately so nothing looks mismatched. The chosen level is written to `~/.hi_jass/settings.json` and restored automatically on the next launch.

1.19 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species s (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s} Z_e^2 Z_s^2 n_e \ln \Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in cm^{-3} . The model uses $Z_e = Z_s = 1$ for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$\tau_E = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$IE = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$IE = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where V is the plasma volume and $e = 1.602176634 \times 10^{-19}$ J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2}V(T_e - T_i)(n_D\bar{\nu}_{ei,D} + n_T\bar{\nu}_{ei,T})(10^3 e)$$

The absolute values make τ_{IE} positive regardless of whether electrons heat ions ($T_e > T_i$) or ions heat electrons ($T_i > T_e$).

1.20 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\text{thermal}} + P_{f,\text{beam}}$$

1.20.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\text{thermal}} = n_{D0}n_{T0}\langle\sigma v\rangle_{DT}(T_i)VE_f$$

Here n_{D0} and n_{T0} are the thermal species densities, V is the plasma volume, and:

$$E_f = 17.6 \text{ MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at T_i , because the relative velocity is that of the thermal ions, not the electrons.

1.20.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\text{beam}} = \sum_j VE_f \int_0^{E_{b,j}} n_{\text{target},j} f_j(E) [\sigma v]_j(E) dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad \int_0^{E_{b,j}} f_j(E) dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E)v_j(E), \quad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

Cross-section energy. The Bosch-Hale cross-section fits take the centre-of-mass energy, while every beam-target and beam-beam call passes the lab energy of a deuteron on a target at rest, $\frac{1}{2}m_D v_{rel}^2$. By default (historical) the fits are evaluated at that lab energy. With `cm_energy_correction` (HOT-Jass: MODELS, "COM energy in beam cross-sections") they use

$$E_{cm} = E \frac{m_{target}}{m_D + m_{target}} : \quad E_{cm} = E/2 \text{ (D-D)}, \quad E_{cm} \simeq 0.6 E \text{ (D-T)}.$$

The lab-energy default over-predicts D-D beam-target rates (about 3x at 60 keV; MAST neutron rates 2-3.5x TRANSP instead of 0.5-0.9x) and misplaces the D-T resonance (beam-target D-T too low above about 80 keV, x0.5 at 120 keV). Thermal reactivities are not affected. Off by default until checked on more cases.

For a D-T beam-target reaction $E_f = 17.6$ MeV. For a D beam, $n_{target,j} = n_{T0}$; for a T beam, $n_{target,j} = n_{D0}$. The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{useful,j} \tau_{S,j}}{E_{b,j} (10^3 e) V}$$

The input shine-through fraction is used to determine captured beam power; the plotted P_{shine} is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating $n_{b0,j}$.

The fusion post-processing always uses the $(1 - \rho^2)$ profile weighting: the thermal contribution is

$$P_{f,thermal} = V E_f \langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT} [T_i(\rho)] \rangle_V,$$

and the beam-target contribution uses the local target density and the local slowing-down distribution evaluated with $T_e(\rho)$. The n_{D0} , n_{T0} , T_e , T_i used here are the **on-axis** values (see *Profile-corrected 0-D*).

1.20.3 D-D fusion

The D-D channel (present whenever the deuterium fraction $x_D > 0$) is evaluated with the Bosch-Hale D-D fits, **branch-resolved**:

branch	products	E_f	neutron
D(d,p)T	T (1.01) + p (3.02 MeV)	4.033 MeV	none
D(d,n) $\bar{3}$ He	$\bar{3}$ He (0.82) + n (2.45 MeV)	3.269 MeV	2.45 MeV

Thermal:

$$P_{f,DD}^{th} = V [E_{ddp} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddp} (T_i) \rangle_V + E_{ddn} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddn} (T_i) \rangle_V],$$

with the $\frac{1}{2}$ identical-particle factor and the same $(1 - \rho^2)$ profile weighting as the D-T thermal term.

Beam-target (a fast D ion on the thermal-D population **no** $\frac{1}{2}$ factor, the two populations are distinct):

$$P_{f,DD}^{\text{beam}} = \sum_{j \in \text{D beams}} V \int_0^{E_{b,j}} n_{D0} f_j(E) [E_{ddp} \sigma_{ddp}(E) + E_{ddn} \sigma_{ddn}(E)] v_j(E) dE.$$

The cross-sections take the **lab-frame** deuteron energy (same convention as σ_{DT}). D-D does **not** produce alphas, so the alpha self-heating fixed-point closes on the D-T power only ($P_\alpha = f_\alpha(3.5/17.6)P_{f,DT}$).

1.20.4 Neutron rate

$$Y_n = \frac{P_{f,DT}}{17.6 \text{ MeV}} + R_{ddn}^{\text{th}} + R_{ddn}^{\text{beam}} (+ Y_n^{\text{bb}}),$$

i.e. one 14.06 MeV neutron per D-T reaction plus the 2.45 MeV neutrons from the D(d,n) branch, plus the beam-beam contribution when that's enabled (see "Beam-beam fusion" above). Named Y_n , not R_n , to avoid confusion with the fast/thermal energy-density ratio R reported elsewhere. Reported on the Dashboard, broken down by channel (beam-target / thermal / beam-beam / full) in the Scan tab's `Y_beam`, `Y_thermal`, `Y_bb`, `Y_full` group and the Summary tab's "Neutron rate" panel, and as a slice of the Fusion tab's pie chart.

1.20.5 Current model limitations

- The beam-target integral uses stationary target ions, so the solved T_i is not included in that reaction integral, and the beam is treated as a single full-energy component (no half/third fractions).
- The tritium and ^3He bred by D-D are not burned further ("cat-DD" is neglected).
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

1.21 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

The T_e, T_i scan plot's y-axis top is $\min(\max(T_e, T_i), 100 \text{ keV})$ – the actual data maximum, unless it exceeds 100 keV (the upper edge of the Bosch-Hale fits' validated range), in which case the axis is capped there so one runaway hot-ion point doesn't wash out the rest of the curve. The bottom is fixed at 0 (not autoscaled) so a fixed top never stretches the axis into an apparent negative margin.

The n_D, n_T, n_{b0} scan group's otherwise-empty 4th panel shows the fast-ion / target-ion density ratio n_{b0}/n_{Target} , where n_{Target} is whichever thermal D-T species the *dominant* (by injected power) NBI beam reacts with in the beam-target channel – n_T for a D beam, n_D for a T beam.

Four further scan groups:

- P_e, P_i, P_{ie}, Q ("Powers") – $Q = P_{f,tot}/P_{NB}$ replaces the flat shine-through power here (moved to "Losses" below, where it belongs with the other loss channels).
- $P_{f,tot}, P_{f,th}, P_{f,b}, P_{f,bb}$ ("Pf") – the previously-empty 4th panel now shows $P_{f,bb}$ (Sec. "Beam-beam fusion" above – a flat zero while that's off), alongside the thermal and beam-target D-T/D-D totals; P_{useful} is still available from the Dashboard.
- $P_{\text{shine}}, P_{\text{orbit}}, P_{\text{cx}}, P_{\text{lost}}$ ("Losses") – the three loss channels (whichever CX-loss model is active feeds P_{cx}) plus their sum $P_{\text{lost}} = P_{NB} - P_{\text{useful}}$.

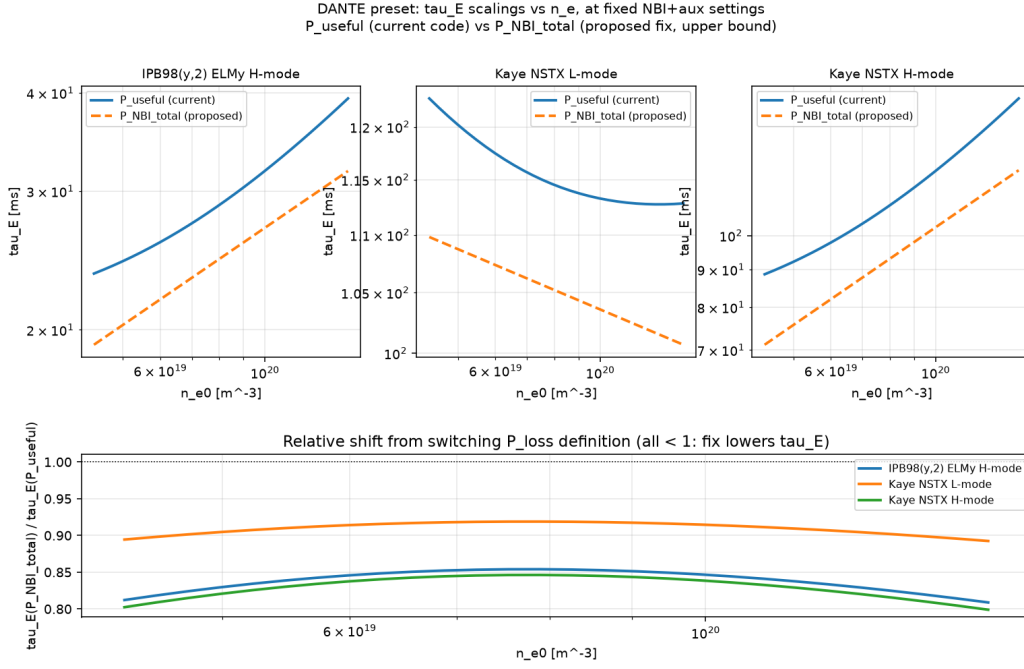
- $Y_{\text{beam}}, Y_{\text{thermal}}, Y_{\text{bb}}, Y_{\text{full}}$ ("Neutrons") – the neutron rate split by reaction channel: beam-target (D-T + D-D), thermal-thermal (D-T + D-D), beam-beam (Sec. "Beam-beam fusion" above – a flat zero while that's off), and the full total ($Y_{\text{beam}} + Y_{\text{thermal}} + Y_{\text{bb}} = Y_{\text{full}}$ exactly). Named Y , not R , so it isn't confused with the $R = u_{\text{fast}}/U_{\text{th}}$ scan group.

1.22 Important notation

- T_e : electron temperature
- T_i : thermal-ion temperature
- P_e : beam power deposited into electrons
- P_i : beam power deposited into ions
- L_e, L_i : beam-power electron / ion heating fractions ($L_e + L_i = 1$)
- P_{ie} : electron-ion energy exchange power
- $P_{\text{ECRH}}, P_{\text{ICRH}}$: ECRH and ICRH auxiliary heating power
- $P_{\text{aux},e}, P_{\text{aux},i}$: auxiliary heating to electrons / ions
- $P_\alpha, P_{\alpha,e}, P_{\alpha,i}$: fusion alpha self-heating power (total, to electrons, to ions)
- $\tau_{E,e}, \tau_{E,i}$: electron / ion energy confinement times used in the balance
- n_{thermal} : thermal-ion density, $n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$
- n_D, n_T : thermal deuterium and tritium densities
- n_{b0} : total fast-ion density
- ρ_{Li} : fast-ion toroidal-field Larmor radius at the injection energy
- Δr : passing drift-orbit radial width, $q_* \rho_{Li}$ (large-aspect model)
- ρ_θ : fast-ion poloidal gyroradius, $(B_t/\bar{B}_p) \rho_{Li}$ (ST orbit models)
- $w_{\text{pass}}, w_{\text{ban}}$: ST circulating / banana orbit widths
- f_t, f_c : trapped / circulating particle fractions (arbitrary A)
- q_a : arbitrary-aspect-ratio edge safety factor (Uckan form)
- f_{orbit} : first-orbit loss fraction of captured beam power
- P_{orbit} : first-orbit loss power
- $P_{f,\text{tot}}, P_{f,\text{th}}, P_{f,b}$: total, thermal, and beam-target fusion power (D-T)
- $P_{f,DT}, P_{f,DD}$: D-T and D-D fusion power ($P_{f,\text{tot}} = P_{f,DT} + P_{f,DD}$)
- R_n : neutron rate (14 MeV D-T + 2.45 MeV D-D branch)
- $\langle T \rangle, T_0$: volume-averaged and on-axis temperature (profile-corrected 0-D)
- p_n, p_{Te}, p_{Ti} : density and electron / ion temperature profile exponents ($(1 - \rho^2)^{2p}$)
- τ_{IE} : effective electron-ion exchange time
- $p_{\text{th}}, p_{\text{fast}}$: thermal and fast-ion pressure
- β_t : toroidal beta
- f_{dil} : fast-ion dilution, $n_{b0}/(n_{\text{thermal}} + n_{b0})$
- τ_p^* : effective particle confinement time (incl. recycling)
- N_s : thermal inventory of species s , $\langle n_s \rangle V$
- $S_{\text{beam} \rightarrow s}, S_{\text{ext},s}, R_{\text{burn},s}$: beam, external fuelling and burn-up particle rates

1.23 Appendix A: confinement-scaling P definition – expected impact

This appendix expands on the caveat noted under "Confinement time" above. It is exploratory analysis, not a validated result: standalone scripts, outside the app and the solver, generated the three figures below (see `docs/Validation/tauE_scaling.Ploss_definition_scan.py` and `tauE_scaling.combined_comparison.py`). **Not yet acted on** – see the "confinement-scaling-p-loss-definition" note if revisiting this.



1.23.1 The question

IPB98(y,2) and the two Kaye NSTX scalings take a power argument, P . HI-Jass currently feeds them $P_{\text{useful}} + P_{\text{aux}} - \text{NBI power after shine-through}$, first-orbit loss, and charge-exchange loss are all subtracted. The experimental databases these scalings were fit to use the coarser convention instead (absorbed power only, i.e. shine-through-corrected but *not* further reduced for orbit-loss/CX-loss of the already-absorbed fast-ion population). HI-Jass's convention therefore under-feeds P into these negative-exponent scalings, which should **overestimate** τ_E whenever orbit-loss/CX-loss are significant – worst for compact/ST-like geometries such as this project's own DANTE default.

1.23.2 DANTE: how much would the fix change τ_E ?

For the DANTE preset, at fixed NBI/ECRH settings, switching from P_{useful} to the raw total injected NBI power (an upper bound on the proposed fix – shine-through would still be subtracted in the real fix) lowers τ_E by **~10-20%**, depending on the scaling and the density:

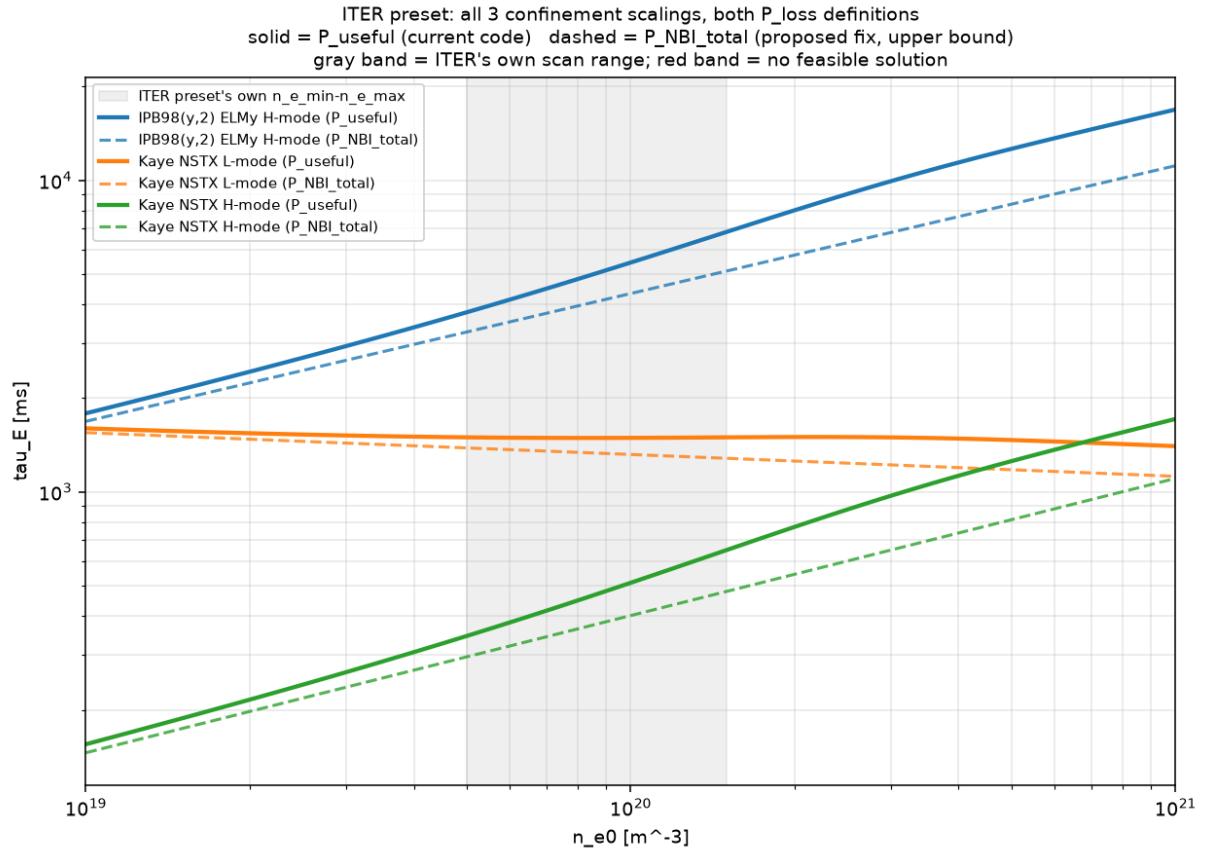
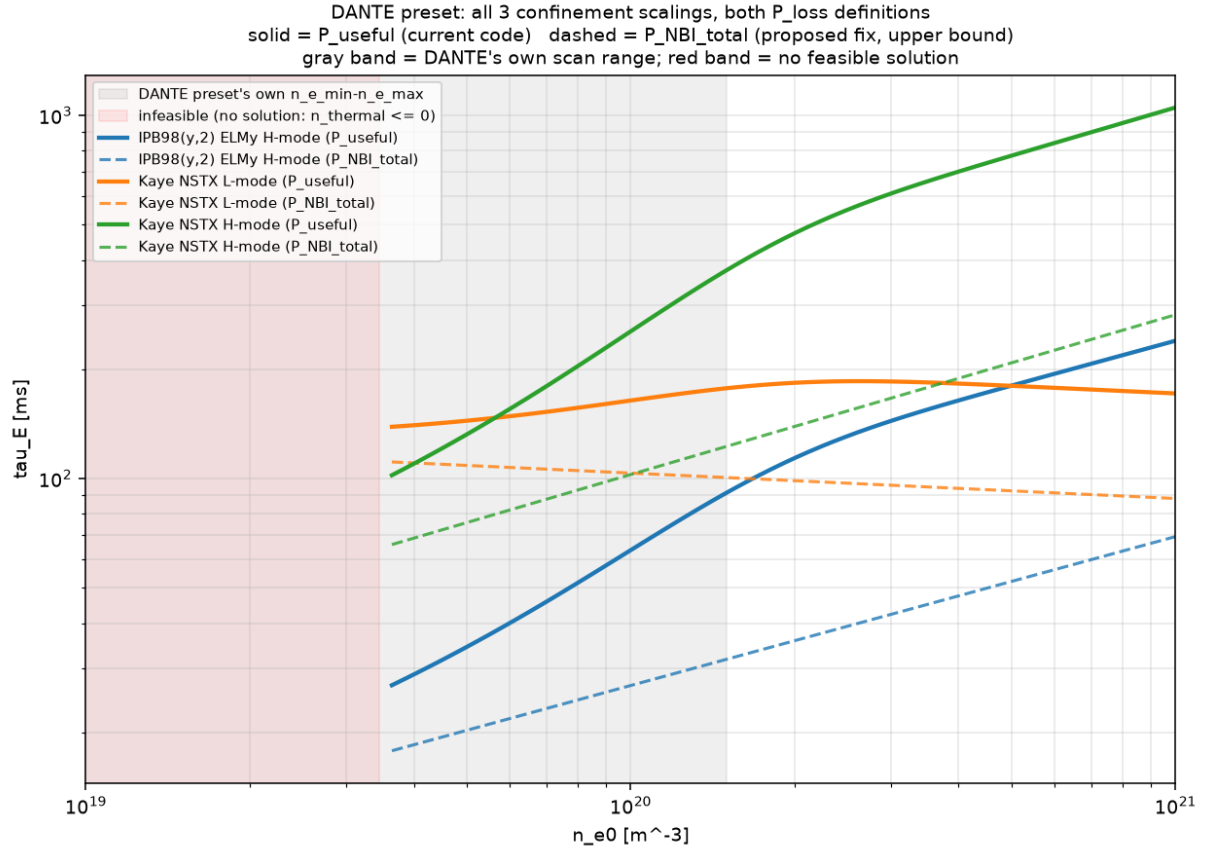
Orbit-loss + CX-loss account for ~25-32% of DANTE's injected NBI power over this scan – that gap is the direct cause of the shift above.

1.23.3 But the scaling choice itself matters far more

The same DANTE comparison, with all three scalings drawn on one axis, shows the **scaling-choice spread is ~4-6x** across the density range – 20-30x larger than the P -definition effect above. Red shading marks densities where HI-Jass has no feasible solution at all for this preset (the model's thermal-ion density would go negative):

The same view for the **ITER** preset shows an even larger spread (~10-12x), but with the *ranking reversed* – IPB98(y,2) now sits far above both Kaye curves, at multi-second τ_E values:

Read this ITER case with caution, not as competing physics estimates. Kaye's scalings



were fit purely to NSTX data ($I_p \lesssim 1.5$ MA, $B_t \lesssim 0.5$ T); ITER's $I_p = 15$ MA / $B_t = 5.3$ T is a huge extrapolation *outside* that database, in the opposite direction from DANTE/NSTX-like devices. IPB98(y,2), by contrast, was fit including larger, more ITER-like devices, so it is the more trustworthy of the two here – the "spread" for ITER is mostly a statement about extrapolation validity, not a genuine disagreement in overlapping-validity physics.

1.23.4 Takeaway

Fixing the P_{loss} definition (sec. "Known limitation" above) is a real, worthwhile correction (~ 10 - 20% on τ_E), but the choice of *which* scaling to use dominates the total uncertainty by an order of magnitude. Neither should be read as validated for a device far outside that scaling's own fitted dataset – check the Assumptions tab's low-aspect extrapolation flag, and prefer Kaye NSTX only for genuinely NSTX-like ($A \lesssim 1.6$, low B_t) geometries.